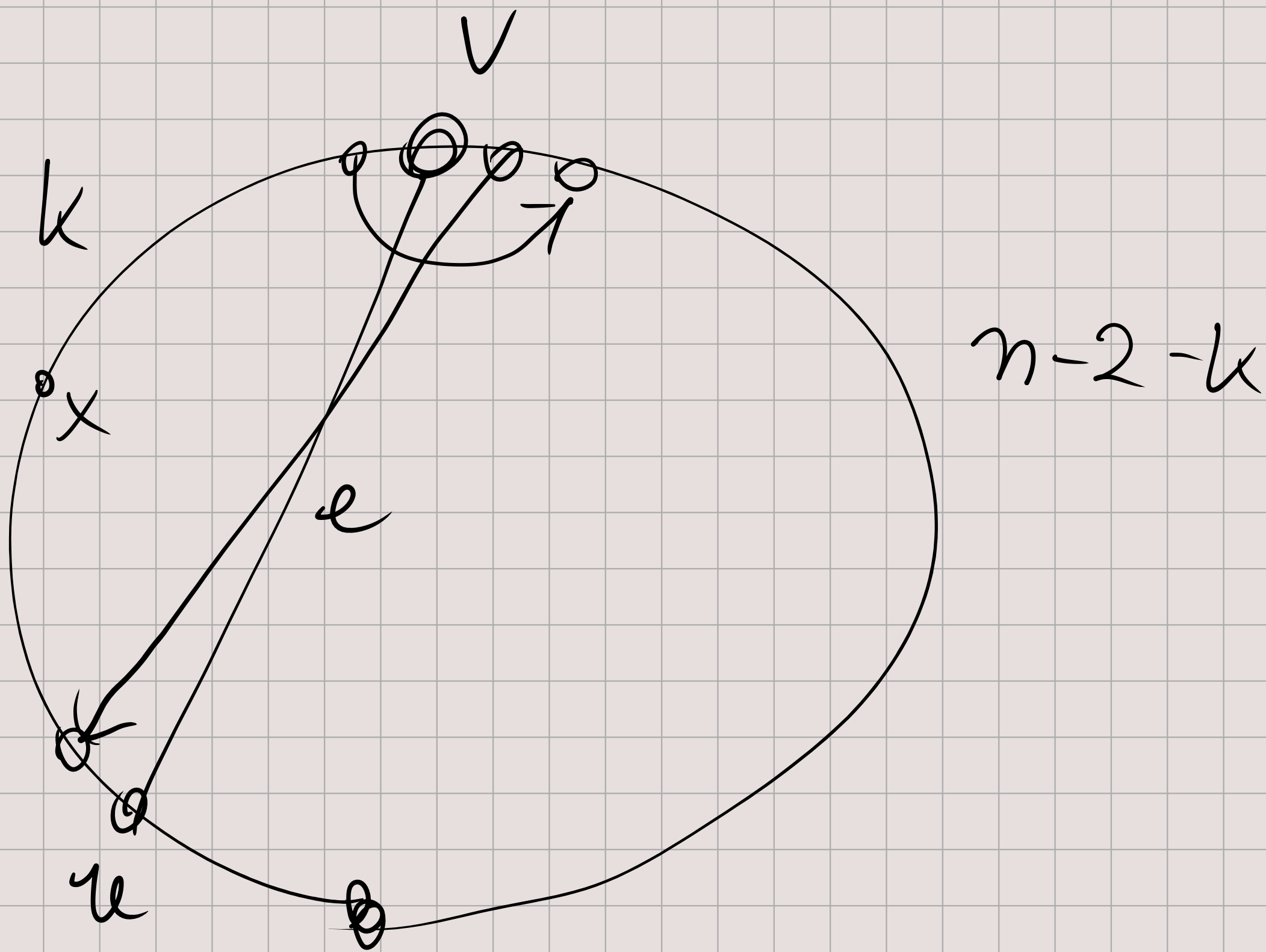
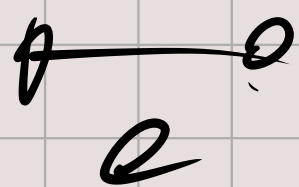




$$k \leq n-2-k$$

$$2k \leq n-2$$

$$k \leq \frac{n-2}{2}$$

$$k \leq \left\lfloor \frac{n-2}{2} \right\rfloor$$

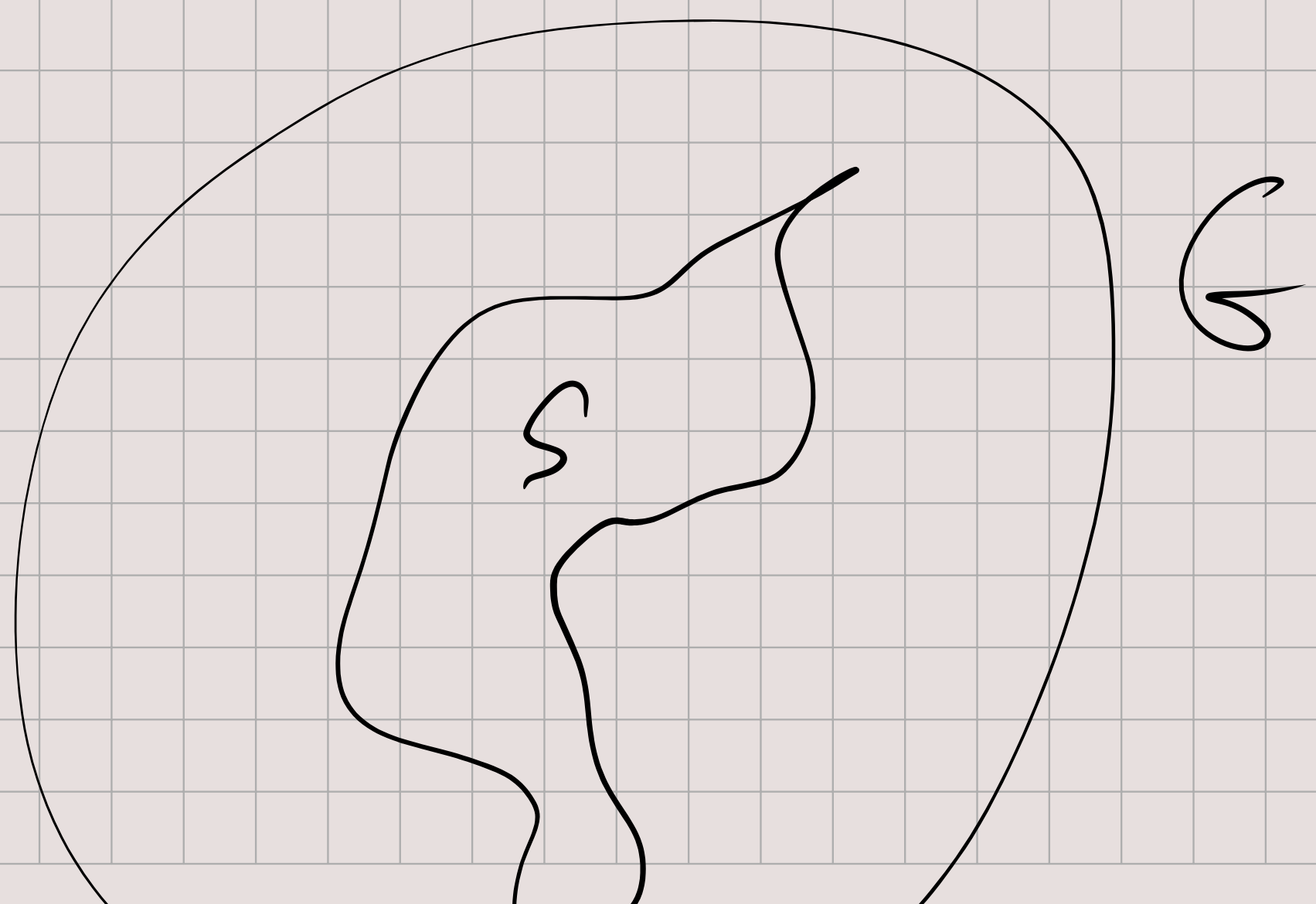
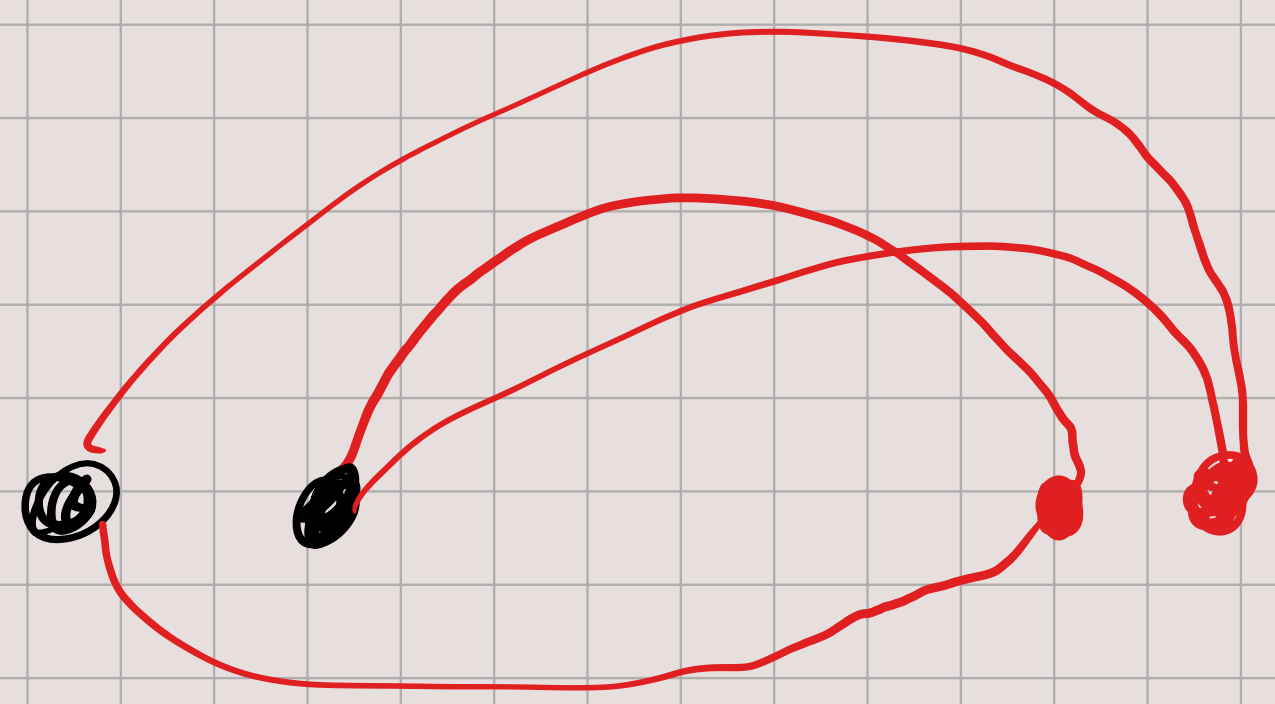
$$\deg_p X \geq \frac{n+1}{2} - k + 1$$

$$n-2 - k \frac{n+3}{2} + k^2 - k$$

$$n-2 - k \frac{n+5}{2} + k^2 \leq 0$$

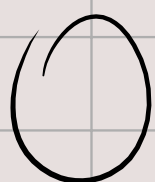
$$n-2 - k \cdot \frac{7}{2} - k \cdot \frac{n-2}{2} + k^2$$

$$arb(G) \leq col(G) - 1 \leq 2 \cdot arb(G) - 1$$

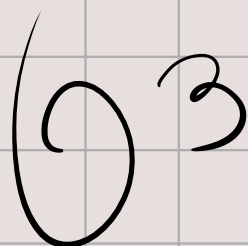
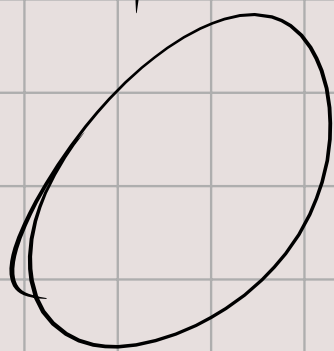


G-S

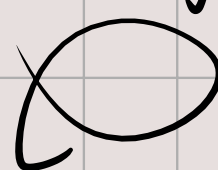
1



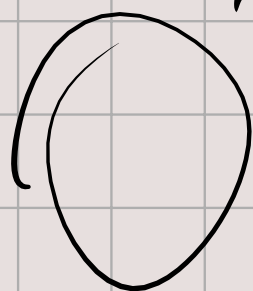
4



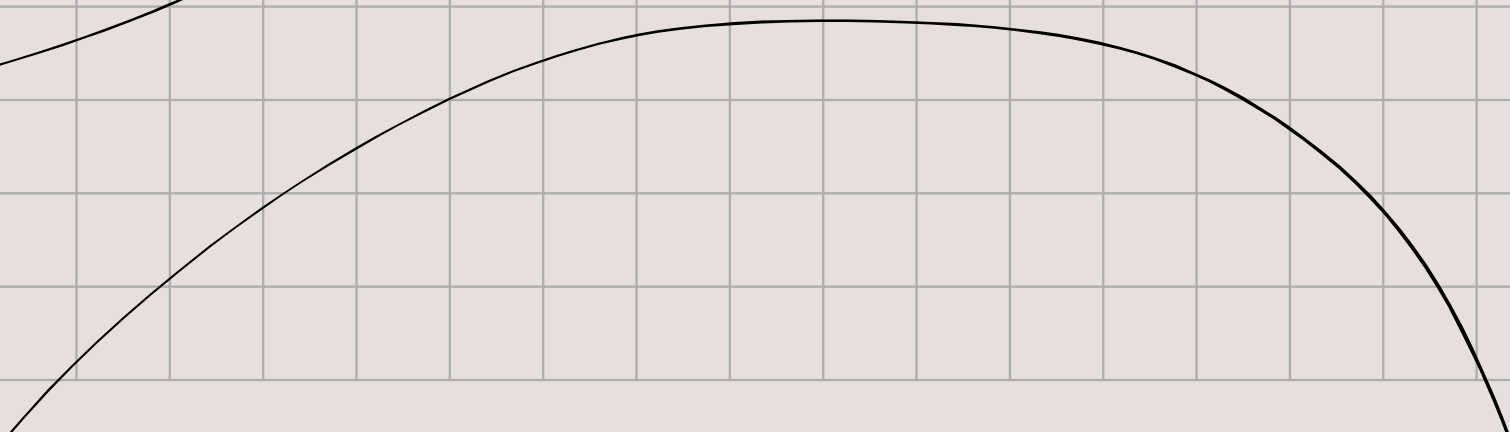
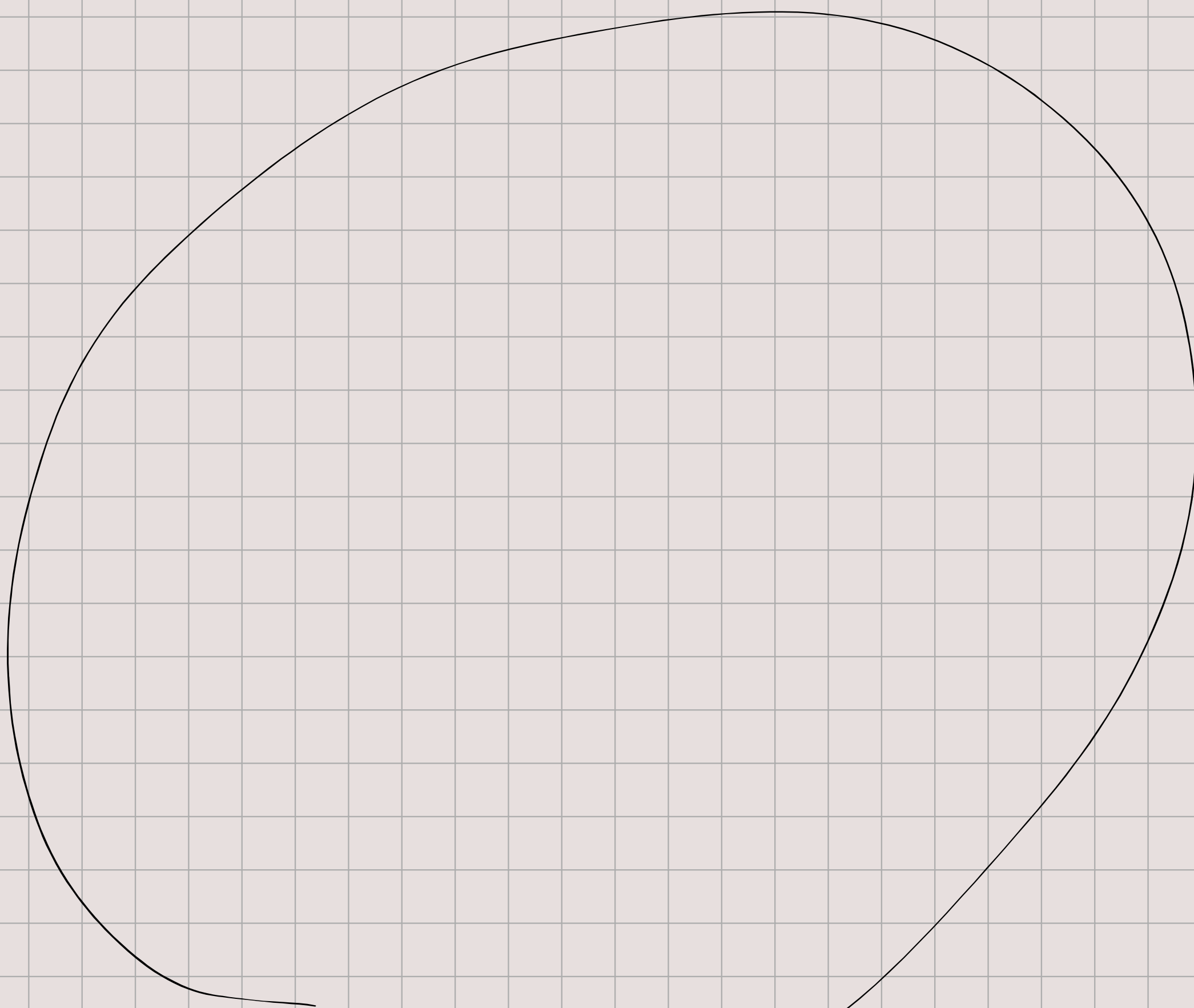
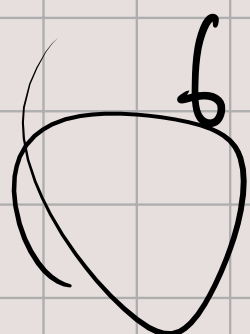
7

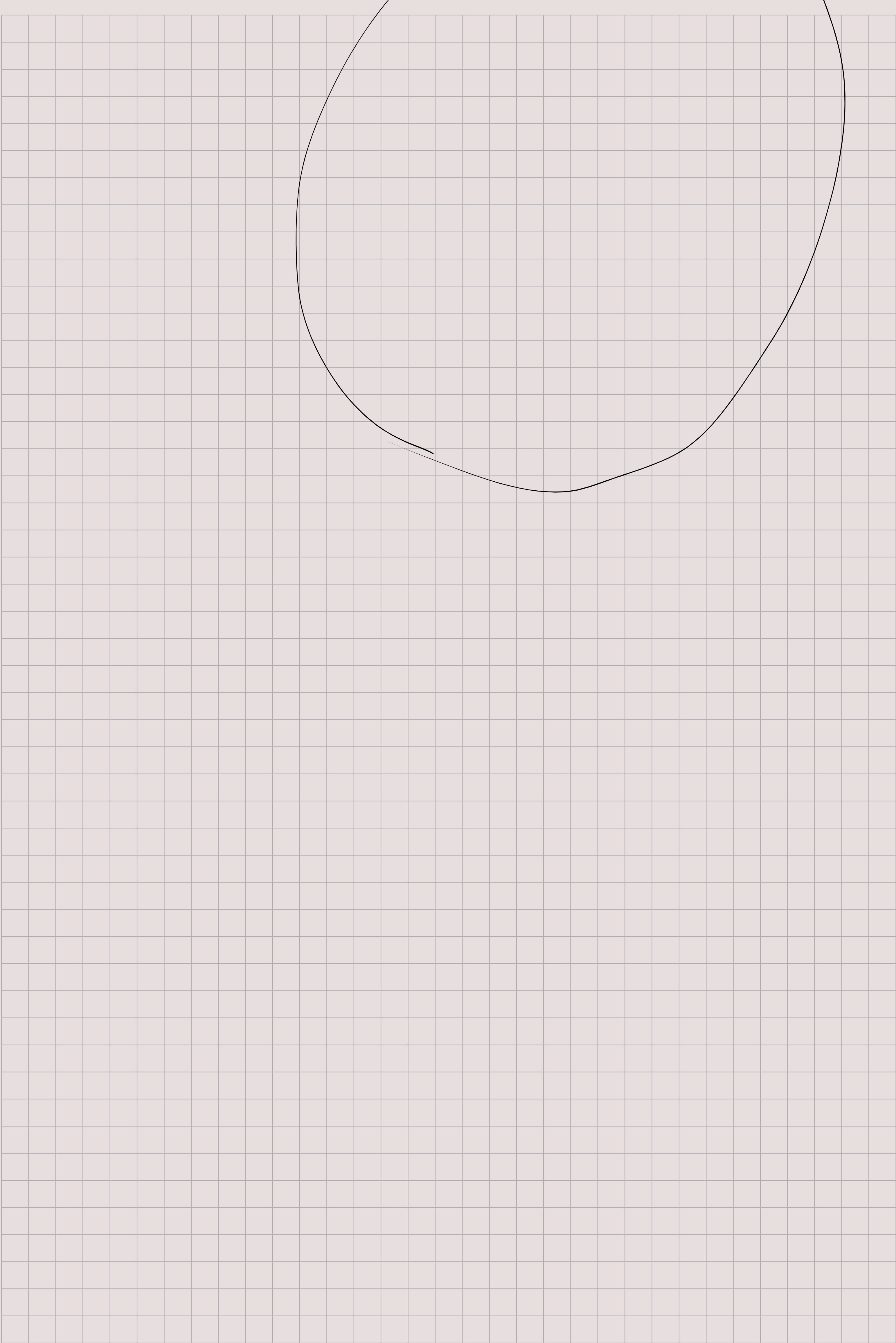


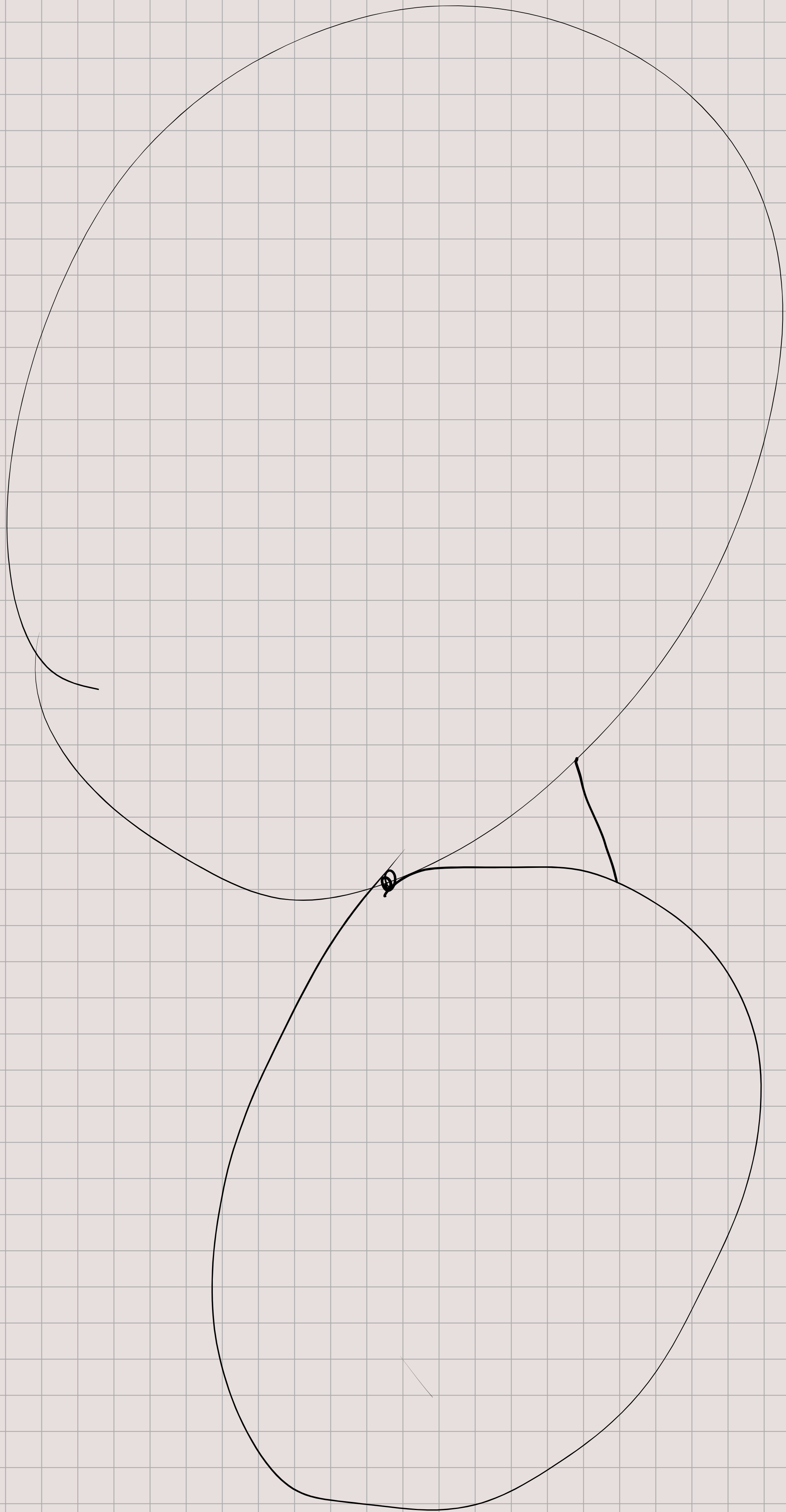
5



6

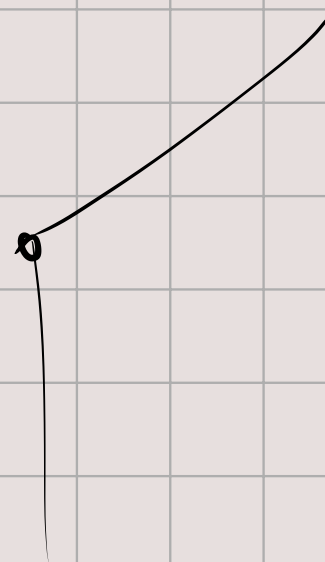
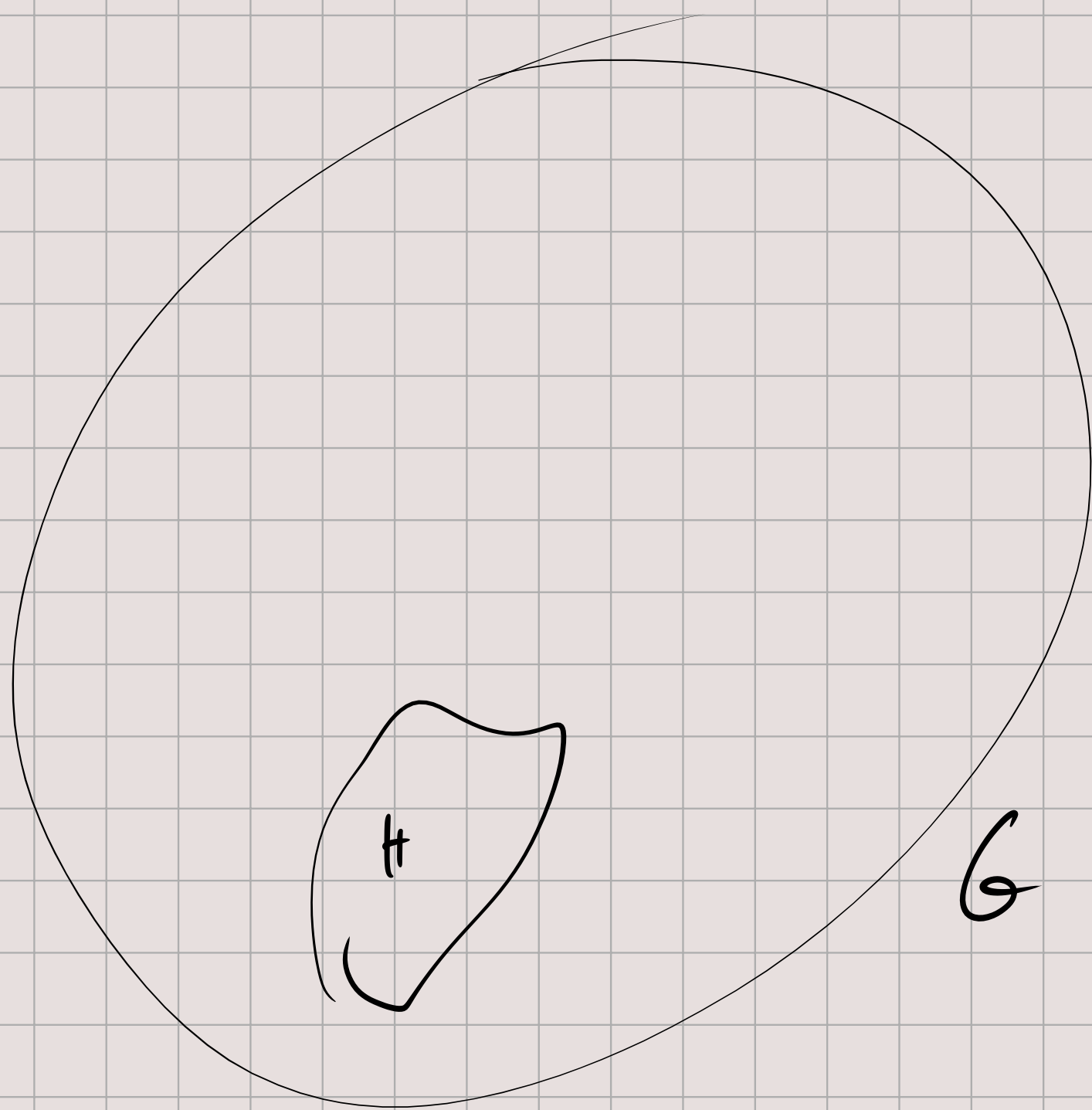






$G - P_4\text{-wolny} \Leftrightarrow \forall \overline{H} \text{ niespójny}$
 $H \subseteq_{\text{ind}} G$
 $H \text{ spójny}$
 $|H| \geq 2$

$\exists \overline{H} \text{ spójny} \Leftrightarrow P_4 \subseteq G$
 $H \subseteq_{\text{ind}} G$
 $H \text{ spójny}$
 $|H| \geq 2$
 $\overline{P}_4 = P_4$



zob

$$\text{col}(G) - 1 = \max \{ \delta(H) \mid H \subseteq_{\text{ind}} G \}$$

$$A_k = \{v \in V(G) \mid \deg_G v \geq 2k(\text{col}(G)-1)\}$$

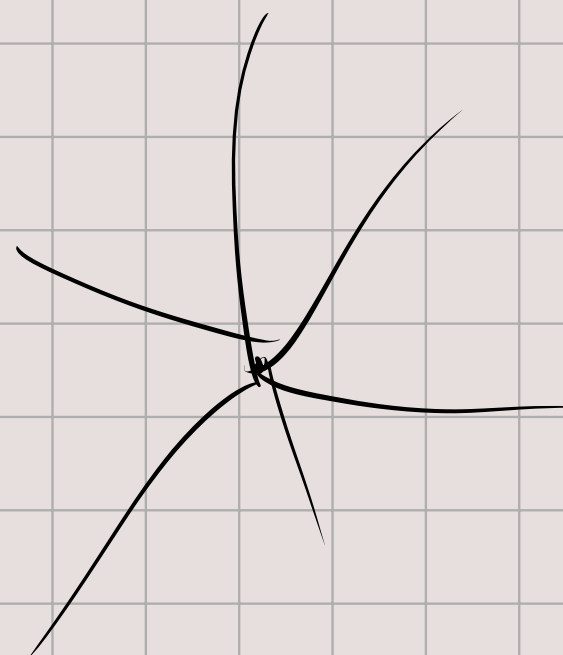
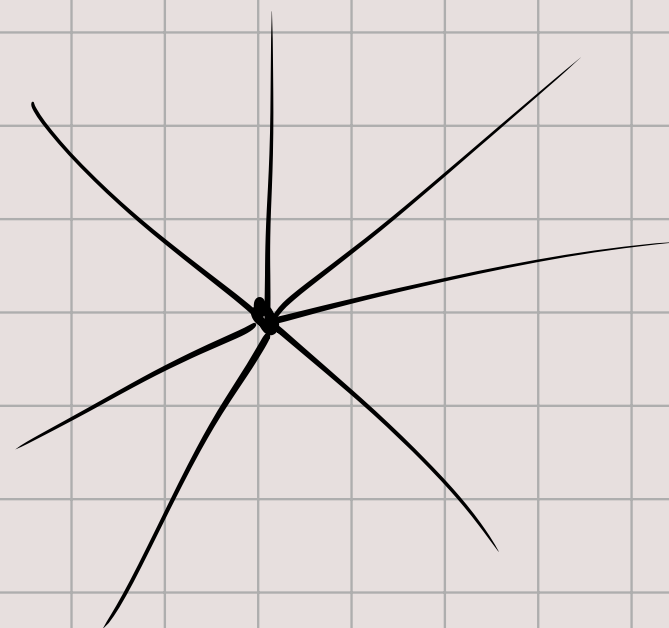
$$\# A_k \leq \frac{n}{k}$$

$$\text{zol} \quad \# A_k > \frac{n}{k}$$

$$A_k \subseteq_{\text{ind}} G$$

$$\delta(A_k) \leq \text{col}(G)-1$$

$$2|E(A_k)| = \sum_{v \in A_k} \deg v > \frac{n}{k} \cdot \delta(A_k)$$



$$e \leq n(\text{col}(G) - 1)$$

